PROJECT PLANNING PHASE

EV Market Analysis Dashboard

Product Backlog, Sprint Planning, User Stories, Story Points, Project Tracker, Velocity and Burndown Chart

1. Project Overview

The project will build an interactive EV Market Analysis Dashboard using the supplied ev_market_train dataset. The workbook contains 200,000 records and 37 columns. The dataset includes vehicle/model details, manufacturer, model year, country, market region, vehicle category, battery and performance specifications, price, safety/customer ratings, CO_2 savings, and annual sales. The dashboard will help users compare EV market patterns through KPIs, charts, filters, and tooltips.

Dataset Snapshot

Item	Details
Dataset file	ev_market_train.csv(1).xlsx
Rows / columns	200,000 rows / 37 columns
Primary measures	annual_sales_units, range_km, price_usd, battery_capacity_kwh, customer_rating, co2_saved_tons_per_year
Main dimensions	manufacturer, category, battery_type, manufacturing_country, market_region, model_year, drivetrain, recommended_use_case
Proposed tool	Tableau Public; spreadsheet for source data inspection; optional Python/Pandas for data quality checks

2. Project Objective and Scope

Objective: create a clear, interactive dashboard that summarizes and compares EV sales, vehicle categories, battery technologies, range, pricing, ratings, manufacturing countries, and model years.

In scope: inspect and clean the supplied data; define consistent measures; create KPI cards and visualizations; add interactive filters/tooltips; validate totals and labels; publish or present the dashboard.

Out of scope: forecasting future sales, causal claims, live data feeds, and external data integration. Findings will describe the supplied dataset only.

3. Product Backlog and Sprint Planning (4 Marks)

Priority: High = required for a working submission; Medium = improves usability; Low = optional polish. Story points are estimates, not hours.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Understanding & Quality	USN-1	As a project team, we can inspect the dataset schema and confirm field meanings and data types.	3	High	Project team
Sprint-1	Data Understanding & Quality	USN-2	As a user, I can rely on key fields being checked for missing values, duplicates, invalid ranges, and inconsistent categories.	5	High	Data analyst
Sprint-1	Data Understanding & Quality	USN-3	As a reviewer, I can see a documented list of data-quality decisions and limitations.	2	Medium	Data analyst
Sprint-1	KPI Design	USN-4	As a user, I can see	3	High	Dashboard

			total recorded annual sales, average range, average price, and average customer rating at a glance.			developer
Sprint-2	Sales Analysis	USN-5	As a user, I can compare recorded annual sales by manufacturer.	3	High	Dashboard developer
Sprint-2	Category & Battery Analysis	USN-6	As a user, I can explore sales by vehicle category and the distribution of battery types.	3	High	Dashboard developer
Sprint-2	Range & Price Analysis	USN-7	As a user, I can compare vehicle range and price and inspect average range across categories.	3	High	Dashboard developer
Sprint-2	Ratings & Geography	USN-8	As a user, I can explore customer ratings, manufacturing countries, and market regions.	3	Medium	Dashboard developer
Sprint-3	Interactivity	USN-9	As a user, I can filter views by manufacturer, category, battery type, country/region, and model year.	5	High	Dashboard developer
Sprint-3	Interactivity	USN-10	As a user, I can use tooltips, legends, and consistent labels to understand chart values.	3	Medium	Dashboard developer
Sprint-3	Validation & Delivery	USN-11	As a reviewer, I can verify chart totals against the source data and review a concise insights summary.	3	High	QA / reviewer
Sprint-3	Validation & Delivery	USN-12	As a presenter, I can publish or demonstrate the final dashboard with a clear title and usage notes.	2	Medium	Project team

Estimated backlog total: 38 story points. Sprint allocation: Sprint-1 = 13 points; Sprint-2 = 12 points; Sprint-3 = 13 points.

4. Sprint Schedule and Estimation

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (Planned)	Sprint Release Date (Actual)
Sprint-1	13	5 days	19 Oct 2026	23 Oct 2026	13	To be updated
Sprint-2	12	5 days	26 Oct 2026	30 Oct 2026	12	To be updated
Sprint-3	13	5 days	02 Nov 2026	06 Nov 2026	13	To be updated

Dates are a proposed plan and should be changed to match the team's actual schedule. "Actual" release dates and completed points must be filled in after each sprint.

5. Project Tracker and Team Velocity (4 Marks)

Assumption for planning: a 10-day sprint with an estimated velocity of 20 story points. Average velocity per day = 20 ÷ 10 = 2 story points/day. The current backlog estimate is 38 points, so the team should plan capacity and adjust scope after tracking actual completion.

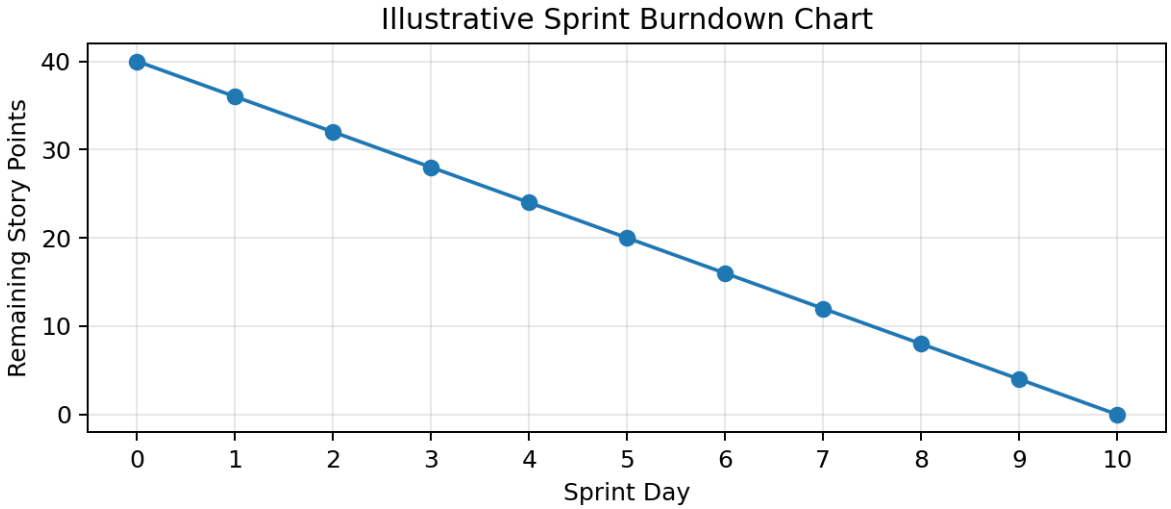
Track at each review: story points committed, story points completed, open tasks, blockers, data-quality issues, dashboard validation status, and release readiness.

Sprint	Committed Points	Completed Points	Blockers / Notes	Status
Sprint-1	13	To update	Schema and data-	Planned

			quality checks	
Sprint-2	12	To update	Build charts and KPIs	Planned
Sprint-3	13	To update	Filters, QA, insights, delivery	Planned

6. Burndown Chart

The chart below is an illustrative baseline for a 40-point, 10-day sprint, matching the example in the planning brief. Replace the planned line with daily remaining-work values once the actual sprint begins.



Daily tracking rule: remaining work = total committed story points − completed story points. Update at the end of each working day.

7. Acceptance Criteria

- Dataset fields and data types are reviewed; missing values, duplicates, and unusual values are documented.
- KPI values and chart aggregations are cross-checked against the source dataset.
- Dashboard includes manufacturer sales, category sales, battery distribution, range/price comparison, ratings, country/region, and model-year views.
- Filters and tooltips work consistently and do not mislead users about aggregation.
- The final dashboard has a clear title, readable labels, an insights summary, and a presentation-ready layout.
- If Tableau Public is used, the team understands that published workbooks are publicly viewable.

8. Risks and Mitigation

Risk	Mitigation
Missing or inconsistent values	Profile columns early; document any exclusions or treatment.
Large dataset slows authoring	Use extracts/aggregations where appropriate and keep worksheets focused.
Misleading comparisons across different vehicle types	Label measures clearly and allow category/region filters.
Time overrun	Prioritize core KPIs and required views before optional formatting.
Public sharing exposes the workbook/data	Review publishing settings and use Tableau Public only if public access is acceptable.

9. Deliverables

Cleaned/validated dataset notes; Tableau workbook/dashboard; KPI and visualization set; interactive filters/tooltips; short insights summary; project planning tracker; sprint burndown updates; final demonstration or presentation.

Note: This plan is based on the uploaded dataset structure (200,000 rows and 37 columns) and the provided Project Planning Template. Story points and dates are proposed estimates, not completed-work claims.